

A Case Study Using Web Objects and COSMIC for Effort Estimation of Web Applications

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Abstract

Some size measures have been proposed in the literature to be employed in estimating Web application development effort, but to date few empirical studies have been undertaken to validate them and compare their effectiveness. To address this issue we have performed an empirical study by using as size measures COSMIC and Web Objects, with the aim of verifying their effectiveness as indicators of development effort. In particular, we have built two effort estimation models by applying the Ordinary Least-Squares regression and by using a dataset of 15 Web applications developed by an Italian software company. The performance of the obtained models have been evaluated by using a dataset of further 4 Web applications developed by the same software company some time after the first 15 Web applications. The results reveal that both Web Objects and COSMIC are good indicators of the development effort.

Index terms: Empirical Studies, Effort estimation, Web applications, Web engineering.

1 Introduction

Estimating the effort required to develop Web applications represents an emerging issue in the field of Web engineering since it can deeply affect the competitiveness of a software company [5] [26] [27] [30] [25] [34] [36] [37] [38]. In the context of traditional software engineering, several widely accepted (model-based) methods to estimate software development effort can be found. These methods rely on a formal approach, involving the use of algorithms that take in input some project factors influencing the development effort (such as the software size) and produce an effort estimate [8]. However, to date, this is not the case for Web applications, whose development effort is usually estimated using non-model based methods, mostly relying on expert judgments. A limitation to the diffusion

of model-based methods comes from the lack of widely-accepted ways to estimate Web application size that represents a key cost factor. Indeed, the size measures conceived and widely accepted for traditional software systems, such as *Function Points (FPs)* [3], are not suited for Web-based ones, since they fail to capture some specific features affecting the effort required for applications [36] [37] [38]. In the last years, some methods have been proposed to address this issue. Among them, *Web Objects* [34] and *COSMIC* [14] are of special interest, since in some preliminary case studies they showed interesting results in predicting Web application development effort [15] [37] [38] and revealed their superior performance over standard *FPs* [38]. However it is widely acknowledged that a software measure can be widely adopted in the industrial domain only if it has been validated through several studies that have proved its usefulness [6] [7] [40]. To date, there is a strong need for further empirical evidence to support these size measures.

In this paper we report on an empirical study aimed at assessing the effectiveness of both *Web Objects* and *COSMIC* as indicators of Web application development effort. As for the estimation technique, we exploited a commonly adopted technique, namely the *Ordinary Least-Squares (OLS) regression*, that is a model-based technique [8].

In order to validate the effort estimation we have performed a validation based on an *hold-out sample of applications*. This is theoretically the best option since the validation set is a totally separate dataset from the dataset used to build the effort estimation models [22]. In particular, in our study the effort estimation models, one for each size measures, were constructed using 15 Web applications (training set) developed by an Italian software company. Each model was then validated exploiting further 4 Web applications (test set) developed some time after the first 15. In other words, the estimated effort for the applications in the test set are determined using the effort estimation models obtained using the applications in the training set.

The rest of the paper is organized as follows. In Section 2 we briefly recall the concepts regarding the employed size

Table 1. Web Objects components

Internal Logical Files	logical, persistent entities maintained by the Web application to store information of interest
External Interface Files	logical, persistent entities that are referenced by the Web application, but are maintained by another software application
External Inputs	logical, elementary business processes that cross into the application boundary to maintain the data on an Internal Logical File, access a Multi-Media File, invoke a Script, access a Link or ensure compliance with user requirements
External Outputs	logical, elementary business processes that result in data leaving the application boundary to meet a user requirements (e.g., reports, screens).
External Queries	logical, elementary business processes that consist of a data "trigger" followed by a retrieval of data that leaves the application boundary (e.g., browsing of data).
Multi-Media Files	physical, persistent entities used by the Web application to generate output in multi-media format.
Web Building Blocks	logical persistent entities used to build the Web applications and automate their functionality
Scripts	logical, persistent entities used by the Web application to link internal files and building blocks together in predefined patterns
Links	logical, persistent entities maintained by the Web application to find links of interest to external applications

measures, the estimation technique, and the evaluation criteria. Section 3 reports on the empirical evaluation carried out by using the two sets of 15 and 4 Web applications. Related work is discussed in Section 4 while the conclusions and future work conclude the paper.

2 Employed size measures, estimation technique, and validation criteria

2.1 Web Objects

Web Objects were proposed by Reifer to measure the size of Web applications [34]. In particular, Reifer adds four new Web-related components to the five function types of the *FPA* (*Function Point Analysis*) method, namely *Multi-media Files*, *Web Building Blocks*, *Scripts*, and *Links*. In Table 1 we recall the definition of the Web Objects components as provided by Reifer in his white paper explaining the counting conventions of the *Web Objects* [35]. He devised such list based on the opinion of experts and by analyzing 64 Web applications from various application domains [35].

To apply the *Web Objects* method, firstly the instances of the nine components are counted, and a complexity (low, average, or high) is associated to them. Then, a weight is associated to each counted instance of the components, by using a calculation worksheet [35]. The total amount of *Web Objects* is obtained according to the weights that are related to the complexity of each component, giving an indication of the size of the considered Web application. For

further details see [35]. No empirical study was performed by Reifer to show the effectiveness of this approach. Successively, *Web Objects* have been employed in two case studies achieving interesting results [37] [38].

2.2 COSMIC

COSMIC is a widely adopted method for sizing software, which has been approved as an international standard by ISO (ISO/IEC 19761:2003). This method was originally conceived for business and real-time applications, but it can be successfully applied in any other context, where software is mainly dominated by large amounts of data movements. Conversely, the method is not suited to measure software mostly characterized by complex algorithms or other specialized and complex rules [13]. The basic idea underlying this approach is that, for many kinds of software, the most of development efforts are devoted to handle data movements from/to the persistent storage and the users. Thus, their number can provide a meaningful sight of the system size (see the document: Guideline for sizing business application Software Using *COSMIC-FFP*, version 1.0, December 2005 [13]). To identify data movements, *COSMIC* requires to define a proper context model for the specific application, bounding the software from its operating environment. Basically each data movement crossing a boundary has to be counted. The method considers four kinds of data movements, i.e., *Entry*, *Exit*, *Read*, and *Write*, and the functional size of the software, in terms of *COSMIC Function Point* [13], is given by the sum of all these data movements.

The application of the *COSMIC* method to size Web systems has been analyzed by some researchers in the last years [15] [24] [36]. The difficulties of applying the *FPA* method for sizing an Internet bank system firstly motivated Rollo to use the *COSMIC* method [36]. The successful application of the method to that system made Rollo confident that the *COSMIC* approach was suited for sizing Web applications [36]. However, he did not present any empirical result supporting his thesis. Successively, two studies investigated *COSMIC* approach to estimate Web application development effort [15] [24], obtaining interesting results.

2.3 OLS regression

OLS regression [32] has been widely and successfully employed in the industrial context and in several researches to estimate software development effort [8]. This holds also for Web applications [5] [15] [27] [30] [38].

OLS regression is a statistical technique that explores the relationship between a dependent variable and one or more independent variables [32], providing a prediction model described by an equation

$$(1) \quad y = b_1x_1 + b_2x_2 + \dots + b_nx_n + c$$

where y is the dependent variable (the effort), $x_1, x_2, \dots, x_n$ are the independent variables (the cost drivers), b_i is the coefficient that represents the amount the variable y changes when the variables x_i changes 1 unit, and c is the intercept. In our empirical study we have exploited the *OLS regression* to obtain linear regression models that use the variable representing the effort (in terms of person/hours) as dependent (namely EFH) and the variable denoting the employed size measure (namely WO , representing *Web Objects*, or CFP , representing *COSMIC*) as independent. Once the prediction model has been constructed, the effort estimation for a new system has been obtained by sizing it in terms of WO (CFP , respectively) and using this value in the equation $EFH = b*WO + c$ ($EFH = b*CFP + c$, respectively).

Moreover, in order to validate the obtained effort estimation models we have exploited a *hold-out validation* approach, thus a validation based on the use of an *hold-out sample of applications* [22]. In particular, we employed two different datasets: the first obtained from Web applications developed by a software company in the past and the second dataset obtained from Web applications developed by the same software company after the ones included in the first set. The first dataset (training set) was used to build the effort estimation models while the second dataset (test set) was used to validate the models.

2.4 Evaluation criteria

The accuracy of the estimations has been assessed using $MMRE$ and $Pred(0.25)$ [12], and boxplots of absolute residuals [28]. In the following we briefly recall the main concepts underlying $MMRE$ and $Pred(0.25)$, and boxplots. The *Magnitude of Relative Error* [12] is defined as

$$(2) \quad MRE = |EFH_{real} - EFH_{pred}| / EFH_{real}$$

where EFH_{real} and EFH_{pred} are the actual and the estimated effort, respectively. MRE has to be calculated for each observation in the dataset. To have a cumulative measure of the error, all the MRE values have been aggregated across all the observations, thus obtaining two measures of the central tendency, known as *Mean of MRE* ($MMRE$), and *Median MRE* ($MdMRE$), where the latter is less sensitive to extreme values [27].

The *Prediction at level 0.25* [12] is defined as

$$(3) \quad Pred(0.25) = k/N$$

where k is the number of observations whose MRE is less than or equal to 0.25, and N is the total number of observations. According to Conte *et al.*, a good effort prediction model should have a $MMRE \leq 0.25$ and $Pred(0.25) \leq 0.75$, meaning that at least 75% of the predicted values should fall within 25% of their actual values [12].

Boxplots are widely employed in data analysis since they provide a quick visual representation to summarize the data, using five values: median, upper and lower quartiles, minimum and maximum values, and outliers [23]. They represent a useful graphical representation to compare the distributions of absolute residuals (i.e., $|EFH_{real} - EFH_{pred}|$) obtained with different estimation models.

Moreover, we have tested the *statistical significance* of the obtained results by using absolute residuals, in order to establish if one of the measures provided better results than the others [23] [27]. In particular, we have performed statistical tests to verify whether the two population of absolute residuals, obtained using CFP and WO , respectively, have identical distributions.

3 Empirical Evaluation

In this section we present and discuss the results of the empirical analysis we have performed.

3.1 Dataset

A crucial aspect in any empirical software engineering study is the availability of reliable project data coming from the industrial world. Indeed, often the lack of publicly available datasets is the main problem in the empirical software engineering field that prevents researchers to effectively and efficiently verify hypothesis and ideas, and to transfer in the industrial world proposals conceived in the academic field.

Data for our empirical study have been provided by an Italian software company, whose core business is the development of enterprise information systems, mainly for local and central government¹. Among its clients, there are also health organizations, research centers, industries, and other public institutions. The company is specialized in the design, development, and management of solutions for Web portals, enterprise intranet/extranet applications (such as Content Management Systems, e-commerce, work-flow managers, etc.), and Geographical Information Systems. It is certified ISO 9001:2000, and it is also a certified partner of Microsoft, Oracle, and ESRI.

This company provided us two sets of 15 and 4 Web applications. The Web applications of the second set were developed by the software company after the first 15. The two sets include e-government, e-banking, Web portals, and Intranet applications. They have been developed by exploiting a wide range of Web-oriented technologies, such as J2EE, ASP.NET, etc.. Oracle has been the commonly adopted DBMS, but also SQL Server, Access and MySQL were employed in some Web applications.

The variables employed in our empirical analysis are: EFH , expressed in terms of person-hours, CFP , expressed

¹For confidentiality reasons, the company name is not reported.

Table 2. Descriptive statistics of EFH, CFP, and WO for the study

Dataset	Var	Obs	Min	Max	Mean	Median	Std. Dev.
I	EFH	15	1176	3712	2677.867	2792	827.115
	CFP	15	264	1022	602.400	611	249.113
	WO	15	465	2258	1464.867	1389	543.986
II	EFH	4	578	2686	2287	2406	451.01
	CFP	4	578	645	610.75	610	29.353
	WO	4	1151	1427	1280	1271	122.352

in terms of *CFPs*, and *WO*, expressed in terms of number of *Web Objects*. Table 2 reports the summary statistics performed for *EFH*, *CFP*, and *WO*, related to the two datasets of 15 (I) and 4 (II) Web applications.

3.2 Results

In this section we report on the estimation results obtained by using the two sets of 15 and 4 Web applications and by exploiting the *OLS regression* as modeling technique and *COSMIC* and *Web Objects* as size measures.

We have performed the *OLS regression* to build effort estimation by using as dataset the first set of 15 Web applications. To this end, we have preliminarily carried out an outlier's examination, in order to remove possible extreme values which may unduly influence the models, and verified the assumptions underlying the *OLS Regression*. The analysis highlighted the absence of outliers. Moreover, following the suggestion of Kitchenham and Mendes [30], we have also analyzed the residuals plot and used Cook's distance. We have not found any influential observations that could threaten the stability of the models.

As for the *OLS Regression* assumptions, we have verified that the relationship between the independent variable and the dependent variable is *linear*; the variance of the residuals is constant for all the values of the independent variable; and the residuals are *normally* distributed. We do not report on this analysis for sake of space.

Table 3 shows the results of the *OLS Regression* carried out with the 15 Web applications to verify the quality of the obtained models. In particular, to determine the goodness of fit of the regression models we have used the Coefficient of Determination, R^2 , which measures the percentage of variation in the dependent variable explained by the independent variable. A high R^2 value is an indication of the goodness of the prediction model. Other useful indicators are the F value and the corresponding p -value (denoted by *Sign F*), whose high and low values, respectively, denote a high degree of confidence for the prediction. Moreover, we have performed a t statistic and determined the p -value and the t -value of the coefficient and the intercept for each model in order to evaluate its statistical significance. p -value pro-

Table 3. The results of the OLS regression analysis with CFP and WO

		Value	Std.Err	t-value	p-value
CFP	Coefficient	3.014	0.386	7.801	0.000
	Intercept	862.281	250.613	3.441	0.004
WO	Coefficient	1.246	0.241	5.162	0.000
	Intercept	851.912	375.814	2.267	0.041
	R^2	Adjusted R^2	Std. Err	F	Sign. F
CFP	0.824	0.810	360.099	60.861	0.000
WO	0.672	0.647	491.513	26.645	0.000

vides the probability that a coefficient of a variable is zero, while t -value can be used to evaluate the importance of the variable for the generated model. A p -value less than 0.05 indicates that we can reject the null hypothesis and the variable is a significant predictor with a confidence of 95%. As for the t -value, a variable is significant if the corresponding t -value is greater than 1.5.

Concerning the regression model obtained with *CFP*, the coefficient 3.014 and the intercept 862.281 are significant at level 0.05, as from the t test. The model is characterized by an high R^2 value (0.824) a high F value and a low p -value, indicating that the prediction is indeed possible with a high degree of confidence. The t -values and p -values for the corresponding coefficient and the intercept present values greater than 1.5 and less than 0.05, respectively.

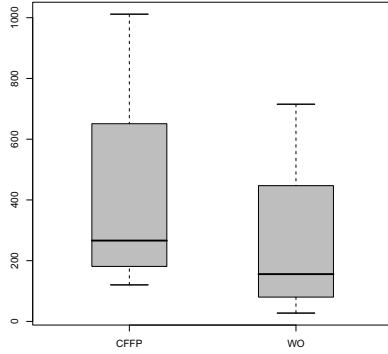
As for the regression model obtained using *WO* as independent variable, the coefficient 1.246 and the intercept 851.912 are significant at level 0.05, as from the t test. This model is characterized by a $R^2=0.672$ that can be considered a fair value, lower than the one characterizing the model based on *CFP* variable (about 0.15). Even if the F value is not so high, the corresponding p -value is less than 0.05. Furthermore, the t -values and p -values for the corresponding coefficient and the intercept present values greater than 1.5 and less than 0.05, respectively, suggesting that they can be considered accurate and significant.

To evaluate the prediction accuracy of the models obtained with *OLS regression* using the first dataset, we have employed as validation set other 4 Web applications (i.e., by performing a hold-out validation [22]). The results are reported in Table 4. Based on the commonly accepted thresholds provided by Conte *et al.* [12], we can conclude that both *CFP* and *WO* are good for estimating the development effort. Thus, for the software company involved in our study, the models obtained by using information from Web applications developed in the past can be profitably exploited to estimate development effort of new applications.

Although the model based on *CFP* has better R^2 and F values than the ones of the model based on *WO* (i.e., the former fits better than the latter), the analysis of boxplots of absolute residuals in Figure 1 shows a better estimation

Table 4. The results of the hold-out validation

	MMRE	MdMRE	Pred(0.25)
CFP	0.22	0.11	0.75
WO	0.14	0.06	0.75

**Figure 1. The boxplots of absolute residuals**

performance of the model constructed using *WO*. Indeed, the median of boxplot for *WO* is more closed to zero than the one of boxplot of *CFP*. Furthermore, the tails and the box length of the *WO* boxplot are slightly smaller than the tails and the box length of the *CFP* boxplot.

As suggested in [23] [27] we have also tested the statistical significance of the results by using paired absolute residuals. The aim was to verify whether there is statistically significant difference in the absolute residuals obtained with the *CFP* and *WO* models, i.e., whether the absolute residuals obtained with the *CFP* model are significant less than those obtained with the *WO* model. To this end, we have performed the *Wilcoxon test*, by considering a 95% confidence interval. This kind of test is used to verify the hypothesis that the mean of the differences in the pairs (of absolute residuals) is zero. The test statistic is the number of positive differences. If the null hypothesis is true, then the number of positive and negative differences should be approximately the same.

The performed test shows that the null hypothesis cannot be rejected, i.e., there is no statistically significant difference for the absolute residuals obtained by using *CFP* and *WO*. Thus, the two measures have comparable prediction accuracies with the employed dataset.

3.3 Threats to Validity

It is widely recognized that several factors can bias the *construct*, *internal*, *external*, and *conclusion* validity of empirical studies. To satisfy construct validity a study has “to

establish correct operational measures for the concepts being studied” [21]. In other words, it represents to what extent the predictor and response variables precisely measure the concepts they claim to measure [27]. A study has internal validity if its outcome can be considered as a function of the measured variables and not of extraneous factors. Conclusion validity is concerned with the ability to draw statistically correct conclusions while external validity is concerned with the ability to generalize the results to other contexts. In the following, we analyze our empirical study with respect to each of these type of validity.

As for the *construct validity*, the choice of the size measures and how to collect them represents the crucial aspects. Regarding the selection of the size measures, we employed two measures that have been used in the past to estimate Web applications development effort [15] [38]. The collection of information about the application measures and actual effort represents the main difficulties to carry out this kind of study. Concerning the effort collection, the software company used timesheets to keep track of this information, where each team member annotates the information about his/her development effort, and weekly each project manager stored the sum of the efforts for the team. In order to collect all the significant information to calculate the values of the size measures, the authors defined a template to be filled in by the project managers. All the project managers were trained on the use of the questionnaires. They took into account the *COSMIC* Measurement manual, version 2.2 [13] and the rules provided in [15] for the collection required to apply *COSMIC* method. Moreover, they employed the counting conventions of *FPA* method [20] and followed the suggestions provided by Reifer in his “Web Objects White Paper” [35] where he explains how to quantify the *Web Objects* components for a given Web application. The projects managers had experience in applying the *FPA* method so they did not manifest any kind of difficulties to collect the information required for our empirical analysis. One of the authors analyzed the filled templates and the analysis and design documents, in order to cross-check the provided information. The same author calculated the values of the size measures.

Some factors should be taken into account for the *internal validity*: subjects’ authoring and designing experience, and reliability of the data and lack of standardization [5] [6] [23] [27]. The subjects involved in the study were professionals who worked in the software company. No initial selection of the subjects was carried out, so no bias has been apparently introduced. Moreover, the Web applications were developed with technologies and methods that subjects had experienced. Consequently, confounding effects from the employed methods and tools can be excluded. As for the reliability of the data and lack of standardization, the questionnaires used were the same for all the applica-

tions and the project managers were instructed on how to use the questionnaires, to correctly provide the required information. Instrumentation effects in general did not occur.

As for the *conclusion validity* we have carefully applied the statistical tests, verifying all the required assumptions. A threat to conclusion validity could be the few number of Web applications composing the dataset. However, it is recognized that getting information on a great number of real world software applications is quite hard [33] [39] and this is especially true for Web applications for several reasons [29]. Several empirical studies are reported in the literature based on few observations (see e.g., [37] [38] which exploit 12 Web applications). Indeed, this kind of study can contribute to provide useful indications that can be further validated in subsequent studies. So other investigations should be performed to verify/confirm the results of our study.

The systems involved in our analysis are representative samples of modern Web applications, taking into account their type, functionalities, target platforms, and complexity. Thus, we think that the type of Web applications did not bias the external validity of the achieved results. On the other hand, it is recognized that the results obtained in an industrial context might not hold in others [9]. Thus, replications of the study taking into account data from other companies is necessary to get a generalization of the results.

4 Related Work

We discuss related work by first taking into account other empirical studies that focused on *COSMIC* and *Web Objects* and then reporting on empirical studies based on different size measures proposed for Web applications.

4.1 Empirical studies with COSMIC and Web Objects

COSMIC was applied in two studies that exploited sets of Web applications developed by students [15] [24], while *Web Objects* was assessed in two studies [37] [38] exploiting an industrial dataset.

Regarding the studies carried out by Ruhe *et al.*, they performed an *OLS* regression analysis based on a dataset of 12 Web applications developed by a medium-size software company operating in Australia and obtained interesting results in terms of prediction accuracy of the method [38]. Moreover, they also used *Web Objects* as size metrics in the context of *WebCOBRA* [37]. In their studies, they obtained $MMRE=0.24$, $MdMRE=0.23$, and $Pred(0.25)=0.67$ for the *OLS* regression model and $MMRE=0.17$, $MdMRE=0.15$, and $Pred(0.25)=0.75$ for the *WebCOBRA* model. Thus, the analysis reported in the present paper has confirmed the goodness of *Web Objects* to estimate effort. Moreover, it is worth noting that our results not only confirm the results

of Ruhe *et al.*, but also extend those results. Indeed, the applications used in our empirical study have larger sizes than the applications used by Ruhe *et al.*. In particular, the mean effort of our datasets is about three times the mean effort of the Ruhe *et al.*'s dataset and the applications used in our empirical study are characterized by a mean number of *Web Objects* about five times the one of the applications used by Ruhe *et al.*. Thus, the *Web Objects* metrics can be considered a scalable measure in the sense that it turns out to be good for applications of larger size, too.

Regarding the empirical study with *COSMIC*, Mendes *et al.* constructed an effort prediction model by applying *OLS* regression analysis and using data on 37 Web hypermedia applications developed by postgraduate and MSc students attending a Hypermedia and Multimedia Systems course at the University of Auckland (NZ) [24]. The derived model did not present reasonable prediction accuracy and replications of the empirical study were highly recommended by the authors. In [15] *COSMIC* method has been applied on 44 Web applications developed by academic students and the model obtained with *OLS* regression analysis was characterized by good estimations if we assume as a reasonable threshold Conte *et al.*'s suggestion [12]. In particular, they obtained $MMRE=0.15$, $MdMRE=0.10$, and $Pred(0.25)=0.75$. However, the scientific literature has often debated on the industrial relevance of results coming from empirical studies with students [5] [11] [27]. Even if in the context of Web application development it is recognized that professional developers are usually young programmers, coming straight from university [27], further analysis exploiting an industrial dataset is needed to verify/confirm the results obtained with the student Web applications. To this end, the empirical studies we present in this paper was carried out using an industrial dataset. The results confirmed those obtained in [15], suggesting that *COSMIC* can be effectively employed to predict Web application development effort.

4.2 Studies on other size measures for Web applications

Besides *COSMIC* and *Web Objects*, other measures proposed in the literature are extensions/adaptations of the *FPS*. In particular, the *OOFPS* method maps the *FPS* concepts into the primitives used in the conceptual modeling phase of *OOWS* [1]. *OOWS* is a method for producing software for the Web. In a recent work [2], a preliminary evaluation of the *OOFPS* has been provided.

Mendes *et al.* proposed several size measures for static and dynamic Web applications, such as number of Web pages, number of graphics, number of features/functionality, etc. [27] [30]. Those measures were used to provide effort estimations, by employing several

techniques, such as Case-Based Reasoning, Linear and Stepwise Regression, and Regression Tree and using Web applications developed by academic students and by companies. Moreover, the results of two surveys and of an empirical study aimed at identifying early size measures and cost drivers for Web cost estimation were presented in [26]. The same authors also presented a survey literature of hypermedia and Web size metrics proposed in recent papers and classifies the analyzed studies according to a proposed taxonomy [25]. The aim was to provide a mean for classifying and understanding the body of knowledge about Web size measures and their employing for effort estimation.

Other two surveys classified Web size measures [10] [17]. Calero *et al.* discussed a wide ranging set of metrics for quantifying Web quality attributes which have been distinguished in three dimensions related to Web Features, Lifecycle Processes, and Quality Characteristics [10]. Dhyani *et al.* presented formulations and comparisons of well-known Web metrics for quantifying Web graph properties, Web page significance, Web page similarity, search and retrieval, usage characterization, and information theoretic properties [17]. However, both surveys did not analyze the use of the proposed size measures for cost estimation.

Baresi and Morasca defined several measures on the basis of attributes obtained from design artifacts, automatically generated with W2000 [4], and conducted an empirical study and two replications for identifying the attributes that may be related to the effort required for designing Web applications (i.e., they did not focus on the total development effort) [5]. These studies involved the students of an advanced university classes on modeling Web applications and the authors think that the designers' background impacted the achieved results and further analysis is required.

Two set of size measures were compared in [16], the first included some length measures (such as number of Web pages, number of server-side script and applications, etc.) while the second contained some functional measures (in particular, the components used to evaluate the *Web Objects*). The results of this study revealed that the first set provided better estimates when using *Case-Based Reasoning* while the second set provided better estimates when using *OLS*. However, the analysis also suggested that there were no significant differences in the estimations and the residuals obtained with the two sets of measures. In [18] the following sets of measures were compared: *Web Objects*, the two sets used in [16], the Tukuruku measures proposed in [31]. The empirical results showed that all the measures provided good predictions in terms of *MMRE*, *MdMRE*, and *Pred(0.25)* statistics and the study largely confirmed the results of previous works. It is worth that the study presented in [24] employed Web applications developed by academic students while the empirical analysis reported in [18] [16] [38] were based on industrial Web applications.

5 Conclusions and lesson learned

We have reported on an empirical study carried out to assess the effectiveness of *Web Objects* and *COSMIC* as indicators of Web application development effort and to compare their prediction accuracy. As for the datasets used in the empirical analysis, we have exploited a set of 15 Web applications developed by a software company to build two effort estimation models, while further 4 Web applications provided by the same software company have been employed to validate the obtained models. In other words, we have performed a validation based on *an independent hold-out sample of applications* [22]. The results have highlighted the usefulness of *Web Objects* and *COSMIC* measures in estimating Web application development effort when employed in combination with *OLS regression*. The analysis has also revealed that the estimation results achieved with these measures are comparable.

Finally, observe that the results presented in this paper have provided some indications for the software company that supplied the data. They might be also relevant for other companies that develop Web applications similar to those used in our investigation. However, a replication of the study, in different setting and with a larger dataset, is required in order to generalize the results to other companies. Indeed, it is widely recognized that several investigations should be performed to verify/confirm empirical results [40]. Thus, as future work, we plan to collect and analyze further data also from other companies. Moreover, we will investigate other estimation techniques (e.g., *Case-Based Reasoning* [39] that have provided interesting results for traditional software systems. As suggested in [19] we also plan to experiment regression models using on one or more data movement types (i.e., *Exit*, *Entry*, *Write*, and *Read*) as independent variables instead of the solely *CFP* value.

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